

Study Notes: Acids, Bases, and the pH Scale

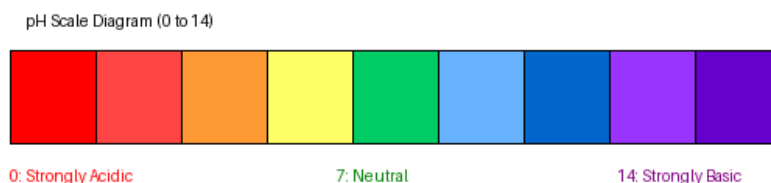
1. Introduction to Acids and Bases

Acids and bases are two classes of chemical compounds that display opposite properties. An acid is a substance that yields hydrogen ions (H^+) when dissolved in water, tastes sour, and turns blue litmus paper red. Common examples include Hydrochloric Acid (HCl) and Citric Acid. A base, on the other hand, yields hydroxide ions (OH^-) in aqueous solution, feels soapy to touch, tastes bitter, and turns red litmus paper blue. Examples include Sodium Hydroxide (NaOH).

2. Understanding the pH Scale

The pH scale is a numeric scale used to specify the acidity or basicity of an aqueous solution. It ranges from 0 to 14. Solutions with a pH less than 7 are acidic, solutions with a pH equal to 7 are neutral (such as pure water), and solutions with a pH greater than 7 are basic or alkaline.

Figure 1: Visual pH Scale Gradient Chart



3. Neutralization Reactions

When an acid reacts with a base, they neutralize each other to produce a salt and water. The general equation for a neutralization reaction is: $\text{Acid} + \text{Base} \rightarrow \text{Salt} + \text{Water}$. For example, $HCl + NaOH \rightarrow NaCl + H_2O$. Neutralization reactions have extensive applications in daily life, such as treating indigestion, neutralizing soil acidity, and treating industrial waste.